



NATIONAL INSTITUTE OF TECHNOLOGY, ROURKELA – 769008
MID – SEMESTER EXAMINATION

SESSION: 2024 – 2025 (Spring)
B. Tech/Dual Degree/Int. MSc./M. Tech. / Ph.D

Subject code: CS 6301 Sub: Foundations of Computer Security
Full Marks: 30

Dept. Code: CS
Duration: 2 hrs

Answer all the questions. All parts of a question should be answered in one place.

Sl. No.	Questions	Marks	CO*	BL**
1.	a) For an affine cipher, we know that the ciphertext 'X' maps to the plaintext 'X' and the ciphertext 'W' maps to the plaintext 'E'. Determine both the decryption and encryption functions. Both answers must be in the form $f(x) = (ax + b) \bmod 26$, where a and b are in between 0 and 25, inclusive. b) If the following key matrix is used to encrypt the plaintext using Hill Cipher then find the corresponding key to be used to decrypt the ciphertext. Decrypt the ciphertext 'CRUAVTLPB'. 12 6 19 3 9 11 4 23 21	[3]	CO1	BL2
		[3]	CO1	BL3
2.	Consider a new version of the Vigenere Cipher that uses a key comprising of a stream of random numbers between 0 and 25 (inclusive). For example, if the key stream is 5 21 3 ..., then the first letter of the plaintext is encrypted with a shift of 5 letters, the second with a shift of 21 letters, the third with a shift of 3 letters and so on. a) Encrypt the plaintext "SEND MORE MONEY" using the key stream 9 0 1 8 9 7 23 14 15 21 11 2 23 b) Decrypt the ciphertext obtained in (a) using an appropriate key stream so that the plaintext obtained is "CASH NOT NEEDED".	[3]	CO1	BL2
		[3]	CO1	BL2
3.	a) Find the multiplicative inverse of the polynomial x^5+x^3+x+1 in $GF(2^8)$ using the irreducible polynomial $x^8+x^6+x^5+x^4+1$. b) Find the multiplicative inverse of 61 in Z_{370}. c) Find the particular and general solution to the following Linear Diophantine equation $19x + 13y = 20$.	[2]	CO2	BL3
		[2]	CO2	BL1
		[2]	CO2	BL2

Sl. No.	Questions	Marks	CO*	BL**
4.	a) Find all sub groups of the group $G = \langle \mathbb{Z}_{17}^*, x \rangle$.	[2]	CO2	BL2
	b) Find the orders of all elements in the group $G = \langle \mathbb{Z}_7^*, x \rangle$.	[2]	CO2	BL2
	c) Show how to multiply $x^3 + x^2 + x + 1$ by $x^2 + 1$ in $GF(2^4)$ over the modulus $x^4 + x^3 + 1$.	[2]	CO2	BL2
5.	a) Explain the phases of DES including the key generation algorithm using suitable diagrams.	[4]	CO3	BL1
	b) Construct a table for the Bifid Cipher with the keyword EFFECTIVENESS and encrypt the plaintext 'fleeatonce'.	[2]	CO2	BL3

*CO: Course Outcome; **BL: Bloom's Taxonomy Level